

RESTORATION ECOLOGY OF INTERMITTENT RIVERS AND EPHEMERAL STREAMS

5.4

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IN A NUTSHELL

- Many intermittent rivers and ephemeral streams (IRES) have been degraded by past and current human disturbances acting on channels and catchments. Ecological restoration seeks to halt the degradation and to build viable self-sustaining systems.
- Well-documented restoration projects focused on IRES are few compared with those on perennial streams.
- Ecological restoration of IRES should initially be planned at the catchment level before the reach level. Restoration should proceed in progressive steps, starting with assessing the disturbance regime, implementing restoration measures and ending with full analysis and reporting on the project outcomes.
- The major form of IRES restoration has focused on revegetating riparian zones. Restoration of stream channels has involved increasing connectivity to allow high flows (floods) to occur unhindered and to allow biota to move. Reinstallation of habitat structure and replenishment of drought refuges have also been restoration targets.
- Ecological restoration of degraded IRES demands urgent attention and action to offset further climate change-induced impacts.

5.4.1 INTRODUCTION

Ecological restoration attempts to return degraded ecological entities (populations, communities, ecosystems) toward their original state or at least to a valued and enduring “self-sustaining” state that displays a number of distinctive attributes relative to degraded ecosystems. Restoration can be guided by concepts, ideas, and empirical results from the discipline of restoration ecology, which in turn relies on principles from ecological theory (Palmer et al., 2006; Lake et al., 2007). Restoration ecology is a relatively new field and while ecological concepts can be, and have been, successfully applied, progress is largely by empirical means (i.e., monitoring, analyzing, and reporting of both successful and unsuccessful restoration projects) and revision of approaches (learning by doing).

In Europe, North America, and Australia, site-based river restoration projects are a major activity for natural resource management groups. However, robust evidence to gauge the outcomes of these projects is scarce (Bernhardt et al., 2005; Brooks and Lake, 2007) despite improvements in the frequency of monitoring (Boulton et al., 2013). Criteria advocated by Hobbs and Norton (1996), Palmer et al. (2005), and Lake et al. (2007) provide a useful guide to maximize the value of restoration projects

(Table 5.4.1). These criteria apply equally to IRES as they do to perennial systems, although available data to guide the assessment may be more difficult to find. Overall, there is a paucity of literature that explicitly addresses the restoration and management of IRES (Fig. 5.4.1).

We begin with a brief outline of the distinctive features of IRES from a restoration perspective. We then summarize the general principles and approaches now adopted in stream restoration, focusing particularly on the importance of understanding the broader landscape processes (both natural and anthropogenic) influencing a stream when assessing and prioritizing restoration actions. After considering the specific types of disturbances and associated restoration actions that are most relevant to IRES,

Table 5.4.1 Restoration steps for IRES, associated issues, and management emphases

Restoration step	Issue	Management emphases
System understanding	Poor characterization of hydrology (particularly at zero flow) Poor characterization of system condition due to dynamism with respect to hydrological state Incomplete understanding of how system originally worked due to limited unimpacted analogs	Installation of depth recorders, mapping of deep pools, longitudinal surveys of channel Requires long-term data collected from degraded and unimpacted systems Compile and synthesize relevant technical literature; develop testable hypotheses about how the system works
Degrading forces	Direct impacts on aquatic habitat and fauna during dry phase Impacts of catchment degradation, exacerbated by ephemerality and hydrological flashiness Impacts of catchment storage on hydrology (timing and duration of zero-flow periods and surface water persistence) Strong interaction between geomorphology and hydrology Limited information about site history	Fencing of channel to prevent livestock and vehicle access, removal of barriers, limit water extraction during zero-flow periods Sediment and nutrient management, revegetation, riparian buffers Farm dam/pond regulation, low-flow by-passes Protect and restore deep pools, prevent livestock access, manage sediment sources by revegetation, consider allowing channel migration Obtain oral histories, historical land surveys, and records
Objective setting	Strong dynamism of biota and ecological functions reflect hydrological phase Strong but gradual cumulative landscape and downstream benefits of site-scale restoration	Set objectives that reflect hydrological dynamism of system (dynamic targets at relevant spatial and temporal scales) Set objectives that recognize downstream and catchment benefits
Management delivery	Relatively more stream length to manage/restore	Management should be planned and undertaken at large scales to maximize cumulative responses and ensure key ecological processes (e.g., connectivity) are restored or maintained
Monitoring	Dynamism of biota and function reflect hydrological phase Responses to restoration may be enhanced, delayed, or modified by hydrological sequences	Indicators should be chosen with respect to hydrological phase: flowing, nonflowing, and dry Long-term monitoring should be anticipated to ensure hydrological phases and sequences

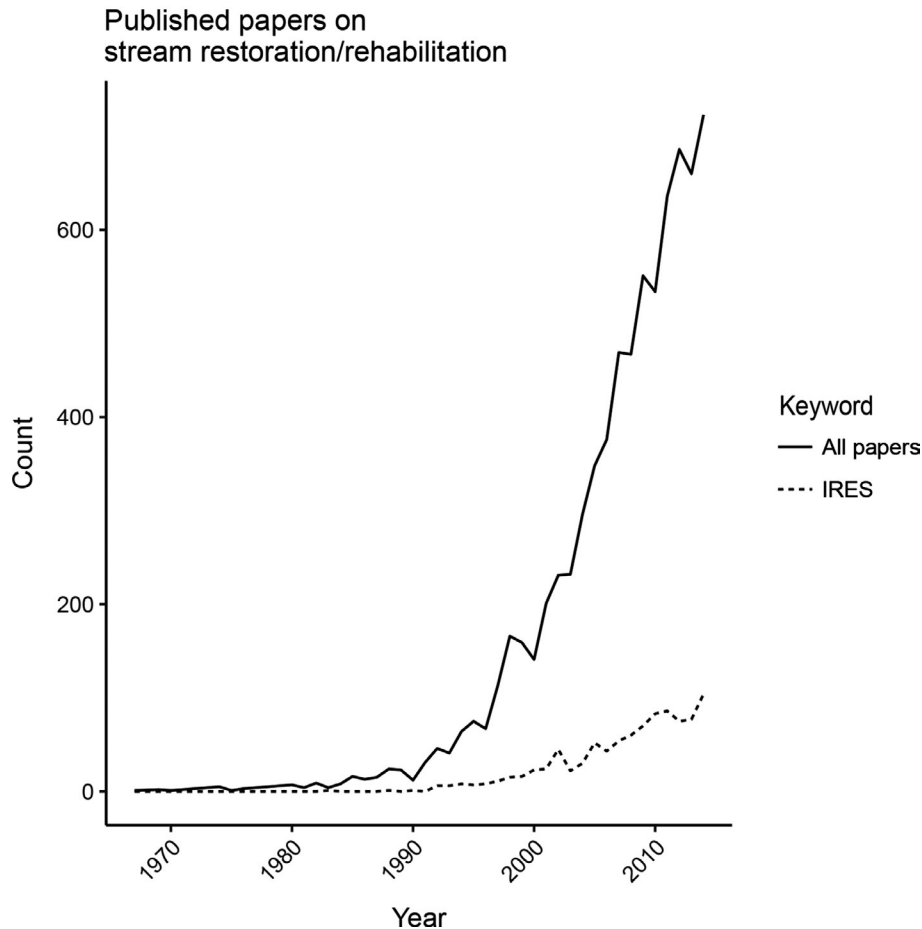


FIG. 5.4.1

Temporal trends in the number of papers published on stream restoration (solid line) and those that refer explicitly to hydrology and IRES-relevant terms more specifically in the title, abstract, and keywords (broken line).

we conclude with suggestions of things to consider now and into the future. Throughout the chapter, we aim to emphasize the importance of using system understanding to guide restoration actions and being aware of potential bottlenecks that may affect system recovery by considering catchment-scale processes potentially influencing restoration trajectories.

5.4.2 IRES AND THEIR CATCHMENTS

Intermittent streams have regular flow periods of differing magnitudes and predictability whereas ephemeral streams have only occasional episodic flows that may include extreme floods. In intermittent streams, flows are either highly variable in volume and timing (e.g., many arid and semiarid

regions, [Levick et al., 2008](#)) or predictably seasonal (e.g., Mediterranean climate regions, [Cooper et al., 2013](#)). Many IRES occur in the headwaters (stream orders 1–3) of perennial systems and can be the crucial water-gathering streams of downstream rivers. For example, in the well-watered Triassic Basins Ecological Region of North Carolina, [Russell et al. \(2015\)](#) located 105 intermittent stream origins in comparison to 52 perennial stream origins, with 35% of stream length being intermittent. The need to conserve headwater streams is widely recognized ([Lowe and Likens, 2005](#)), mainly for the supply of good quality water. Water supply areas often provide the protection for headwater streams that is often lacking in areas cleared for farming.

Flow in streams can occur along three spatial dimensions—longitudinal, lateral, and vertical—and along the critical dimension of time ([Ward, 1989](#); [Chapter 2.3](#)). In IRES, the duration of flow is critical. When flows occur, avenues of connectivity of biota, sediments, organic matter, nutrients, and trace elements are created, and their strength, directionality, and duration fluctuate with changes in flow ([Larned et al., 2010](#)).

In arid and semiarid regions, ephemeral streams can dominate ([Levick et al., 2008](#)). These streams are characterized by long dry periods interrupted by sudden episodic floods when heavy rain falls on their dry, impervious catchments. When flow occurs in ephemeral streams, the major axis of flow is longitudinal. If the channel is shallow and relatively unrestricted, there may be channel widening ([Friedman and Lee, 2002](#)). Vertical flow to the ground water is usually very limited, although it may occur when major floods occur in quick succession ([Baillie et al., 2007](#)). Connectivity involving biota and ecological processes is generally limited in ephemeral streams, and periodic opportunities for dispersal during floods can be critical to biotic persistence ([Chapter 4.5](#)). The condition of the catchment largely controls the amount and type of materials exported downstream during flash floods ([Steward et al., 2012](#)). Floods may carry large amounts of sediments, nutrients, and organic matter and, in the process, reshape channel morphology ([Bull, 1997](#); [Levick et al., 2008](#); [Chapter 2.1](#)). These flows can have considerable transmission losses as they move downstream and may even cease flowing downstream ([Chapter 2.3](#)).

Intermittent streams usually have clearly delimited channels and flows span a range of regimes from irregular to regularly seasonal. Like ephemeral streams, intermittent ones when they flow have longitudinal connectivity, but they differ in having both lateral and vertical connectivity. As flows build, water may move sideways into the parafluvial margins; further increases in flow may move water out of the channel and into and over the riparian zone. With water in the channel, the hyporheic zone can become activated, linking surface water with the groundwater table. The hyporheic zone with its specialized biota and range of biogeochemical processes can exert a strong influence on the ecological dynamics of the channel biota ([Boulton et al., 1998](#)).

Riparian zones in ephemeral streams in arid and semiarid regions may be limited in species composition, cover, and spatial extent. Nevertheless, in arid areas riparian habitats can have a strong influence on the distribution and abundance of biota ([Chapter 4.6](#)) and on ecological processes ([Shaw and Cooper, 2008](#)). As channel flow and groundwater availability become more substantial in intermittent streams, riparian zones become more robust and productive. Even though they may represent only a small percentage of the catchment area, they may contain high levels of biodiversity compared to the catchment as a whole.

5.4.3 STREAM RESTORATION BACKGROUND

The fields of both restoration ecology and stream restoration have evolved considerably over the last few decades, with a considerable shift in emphasis away from thinking about restoration at individual sites to thinking about individual sites within a broader landscape/catchment context (e.g., [Fig. 5.4.2](#);

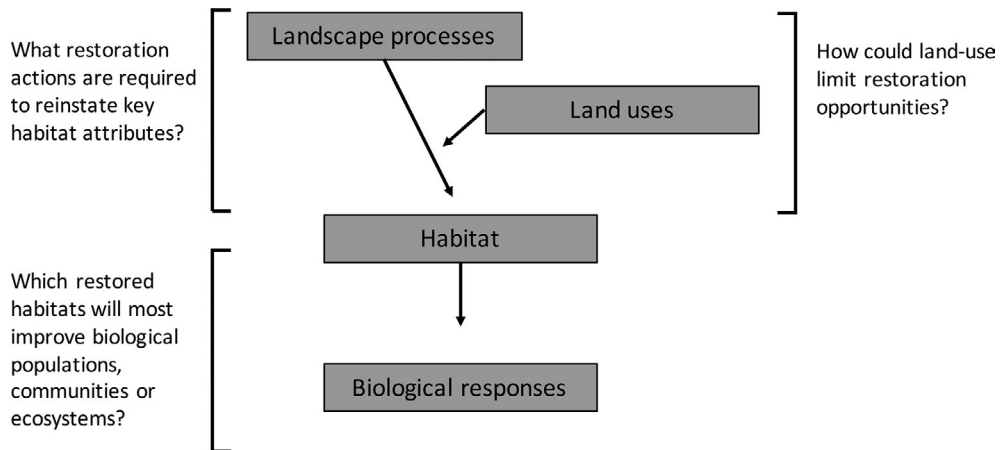


FIG. 5.4.2

Diagram highlighting the role of a landscape perspective in deriving system understanding and planning restoration activities.

Modified from Beechie, T.J., Pess, G., Roni, P., Giannico, G., 2008. Setting river restoration priorities: A review of approaches and a general protocol for identifying and prioritizing actions. N. Am. J. Fish. Manag. 28, 891–905.

Bond and Lake, 2003a; Beechie et al., 2008, 2010). To a large degree, this shift in emphasis has arisen from the recognized failings from approaching restoration within a narrow “site-specific” perspective. A good example of this shift comes in managing restoration of salmonid habitat in streams of the Pacific North West impacted by logging. Here, there has been a gradual evolution from a focus on recreating local habitat-bearing log-jams to restoring connectivity between high-quality habitats and the processes that support natural habitat regeneration, and finally to considering off-shore drivers affecting the population dynamics of diadromous fishes (e.g., Roni et al., 2002; Beechie et al., 2008, 2010). Such shifts in thinking are equally relevant to IRES where connectivity with perennial streams can be critical, and many of the degrading forces and factors that can constrain recovery occur at landscape scales.

Arguably, one of the most important steps in carrying out successful restoration is to first compile a detailed understanding of the stream and its catchment, including the physical and biological processes that operated in the past, operate currently, and are necessary to support a more desirable state (Table 5.4.1; Fig. 5.4.2; Beechie et al., 2008, 2010). This involves gathering historical and contemporary information on climate; geology; catchment geomorphology; vegetation cover; land use; surface and groundwater hydrology and water quality; and the nature, localities, and historical sequences of human settlement and land-use change (e.g., Beller et al., 2016). Stream characteristics to assess include channel morphology and substrata, extent and condition of riparian zones, hydrology (both volumes and timing of flowing and nonflowing periods), and human interventions (e.g., dams, weirs, channel works, bridges).

A vital but often neglected step is to understand the historical sequence of disturbances leading to degradation (e.g., Mika et al., 2010; Beller et al., 2016). This may include the interactions between natural disturbances and the degradation from both legacy (e.g., gold dredging, bushfires) (Bain et al., 2012) and contemporary disturbances (e.g., grazing, water extraction, invasive species). For example,

the creation of arroyos (eroded gulches with steep sides) in IRES of southwestern United States largely occurred from the 1880s to the 1920s with the advent of widespread livestock grazing and frequent floods (Turner et al., 2003; Webb and Leake, 2006). With the cutting of the arroyos, groundwater levels subsequently dropped, threatening the integrity of the riparian vegetation. Attempts must be made to determine the forms of degradation created by the various anthropogenic and natural disturbances and their interactions. In many cases, this may be difficult, largely due to poor historical records. Nevertheless, the historical and current regime of disturbances should be estimated, if not understood, to determine appropriate restoration strategies (Roni et al., 2002). If this is not attempted, then further disturbances or mixtures of disturbances may compromise the restoration effort. This step and subsequent steps in the restoration sequence are summarized in Table 5.4.1.

With an understanding, however incomplete, of the stream and its catchment and of the disturbance regime, an attempt should then be made to devise an overall vision or guiding image of the restored stream and its catchment (Palmer et al., 2005; Mika et al., 2010). Such a vision of the restored condition is essential to guide site selection and types of restorative interventions to achieve project objectives.

The selection of the location and types of interventions that are required depends largely on spatio-temporal scaling. If the restorative intervention is carried out at a large spatial extent (e.g., catchment), then the overall effects will take a long time to eventuate. In contrast, restoration efforts at small spatial extents (e.g., isolated sites) may be relatively rapid to implement and the planned effects may emerge in the short term. If the aim of restoration is to remove a single structure (e.g., dam, barrier) or remove point pollution sources, then the intervention is local, albeit with large-scale positive downstream effects along the stream. However, in many cases the disturbances creating the degradation may affect the whole catchment, even though restoration activities may be occurring at more local scales (Bond and Lake, 2003a; Kondolf and Podolak, 2014). This spatial mismatch may mean that only modest results can be expected from restoration and that prior planning must account for the influence of catchment and landscape processes in deciding what, where, and when to restore (Beechie et al., 2008). This is especially true in the case of catchment disturbances affecting IRES because of the often diffuse and dynamic nature of the terrestrial-aquatic boundary (Chapters 2.3 and 4.9).

After selection of the area or areas of a catchment to be restored, quantifiable goals/targets need to be set. One goal may be to attain the reference condition determined by the conditions in relatively undisturbed streams of the same stream type. For IRES, there may be considerably less published information or primary data to help define targets, due to the historical tendency to focus research on perennial streams in many parts of the world (although there are regional exceptions, Chapter 1). While historical data may be obtained on geomorphology and catchment land use, data describing the hydrology, biota, and ecosystem processes may be limited. The lack of hydrological data for IRES is particularly notable. This occurs because far fewer gauges have historically been located on IRES compared with perennial streams (Chapter 2.2) because IRES often contribute less to catchment water yields. Furthermore, even with gauging data, when cease-to-flow periods occur, it is difficult to determine hydrological attributes, such as the nature of the groundwater system and the distribution and quantity of persistent wetted sections of stream, which provide critical refuge habitats (Jaeger and Olden, 2012). As a result, there is often very poor information on the current and historical flow regimes and the characteristics of IRES to use in understanding their ecology and guiding objective setting. This is further compounded by inadequate descriptions in published ecological work of key hydrological parameters (Fig. 5.4.1), making it difficult to compare across studies.

Despite these challenges, we implore practitioners to try and piece together this historical perspective as there is an obvious and vital role for historical ecology in guiding restoration interventions (Wohl, 2005; Beller et al., 2016). Nontechnical accounts can be valuable, such as the notes from early explorers and from newspaper clippings, both of which can help create a picture of the historical landscape or mention biota with known habitat requirements that may no longer be present. The latter is particularly true for species targeted by recreational anglers and hunters. Photographs, especially “repeats” from the same place, can be valuable for assessing geomorphological and vegetation changes (Turner et al., 2003). With this knowledge from both scientists and informed locals, a plan of expectations could be compiled for the various components and goals set for both flowing and dry periods.

In many cases, due to the strength and forms of legacy and current disturbances, a vision based on reference conditions is infeasible. Instead, it may be necessary to accept irreversible changes and plan for restoration to achieve a novel state (Hobbs et al., 2009). Planning for such a state should be ecologically based and not simply a managerial compromise. In IRES, major components to consider in planning include the riparian vegetation (Stromberg et al., 2009) and fish (Moyle, 2014), the latter often being underappreciated as inhabitants of IRES (Chapter 4.5).

Finally, to assess progress, indicators must be selected. Ideally, they are selected and monitored long before the restorative intervention to provide before-and-after signals of change. Ideal indicators should be inexpensive, easy to measure, and capable of providing a reliable indication of change due to the restoration measure or measures. The feasibility of each of these characteristics will vary by region and with the objectives being measured. It is also important to distinguish between the indicators of intervention measures (e.g., environmental flows) and the indicators responding to the intervention. Some indicators, such as riparian vegetation and channel form, may be sampled at regular intervals while others may need to be monitored on particular occasions (event-based) such as when there is water in the stream (e.g., water quality, aquatic biota; Table 5.4.1, Chapter 5.1).

5.4.4 DEGRADATION AND RESTORATION IN A CATCHMENT CONTEXT

Currently, there is a strong push to examine degradation and recovery processes, and hence restoration activities, from a landscape or whole-of-catchment perspective (Fig. 5.4.2). A major challenge in assessing IRES is discerning where their channels and catchment areas begin, which in their uppermost reaches are often poorly defined. In addition, groundwater features such as recharge areas and aquifers are rarely well mapped, even though they may be critical to streamflow. Consequently, many IRES headwaters are poorly protected by legislation due to inadequate mapping and, in some cases, a lack of concern for IRES (e.g., Svec et al., 2005; Fritz et al., 2013; Chapter 5.2). For example, ephemeral streams are not considered in the US Clean Water Act (O'Connor et al., 2014). Given this limited protection and as their catchments are exploited by humans for a range of uses (Chapter 5.1), the spatial extent and strength of anthropogenic disturbances affecting IRES can be formidable (e.g., Levick et al., 2008; Cooper et al., 2013). Most of the specific threats overlap with those that affect perennial waterways, but the magnitude of impact in many cases can be more severe. Anthropogenic drivers of degradation include livestock grazing, land clearing, mining, timber harvesting, groundwater withdrawal, streamflow diversion, channelization, urbanization, cropping, roads and road construction, off-road vehicle use, camping, hiking, and vegetation conversion (Levick et al., 2008).

Catchment-wide disturbances can be climatic (e.g., floods, droughts, heat waves) or anthropogenic (e.g., land clearing, urbanization, invasive species); in many cases, mixtures of these disturbing forces interact and co-occur (Cooper et al., 2013). Wild fires can damage vegetation cover, including riparian vegetation (Verkaik et al., 2015; Douglas et al., 2015), triggering erosion that can leave a considerable legacy of degradation in affected streams (Reich and Lake, 2015; Verkaik et al., 2015). Similarly, past grazing and floods have left the legacy of arroyos in streams of southwestern United States (Turner et al., 2003; Webb and Leake, 2006). There can also be large-scale disturbances emanating from specific sites that exert strong degrading effects downstream. Examples include point-source pollution from mining operations with widespread toxic effects downstream (Lottermoser, 2003) and dams below which flow may be strongly regulated and/or reduced (Kondolf et al., 2013). In many cases, flow regulation co-occurs with agriculture and pollution (e.g., Cooper et al., 2013; Theodoropoulos et al., 2015), requiring effective restoration to address multiple sources of disturbance. At the subcatchment scale, these may include disturbances such as heavy sedimentation from upstream sources, disruption of riparian zones, stream channelization, and salinization (Cooper et al., 2013).

In urbanized landscapes, the piping and filling of streams (i.e., stream burial) is an extreme form of modification that often affects IRES (Roy et al., 2009). For example, in Baltimore City, Maryland, 66% of all stream reaches have been buried (Elmore and Kaushal, 2008). These low-order IRES are typically areas of high biological activity in undisturbed systems (Meyer and Wallace, 2001). Piping of headwaters causes downstream impacts via increased flow velocities, altered carbon and nutrient inputs, and amplified nitrogen transport; these effects can be exacerbated by increased climatic variability (Kaushal et al., 2008). In the United States, there is particular interest in understanding how headwater degradation affects downstream waters because of important ramifications for regulation under the US Clean Water Act (Leibowitz et al., 2008; Chapter 5.3).

Extraction of groundwater by bores, wells, and pumping from the channel can lower the groundwater table, leading to a loss of flow and increases in cease-to-flow events (Boulton and Hancock, 2006; Wohl et al., 2009). In addition, increasing catchment plant cover (e.g., from grassland to tree plantations, Adelana et al., 2015) can lower the groundwater table and decrease streamflow. With time, reducing groundwater extraction from catchments can elevate the groundwater table and restore streamflow and riparian vegetation (Katz et al., 2009). Another form of a severe catchment-scale disturbance affecting IRES is dryland salinity caused by large-scale clearing of native vegetation. In southwestern Western Australia, for example, the replacement of deep-rooted native vegetation with shallow-rooted crops has led to rising saline groundwater tables, which can then salinize the IRES (Halse et al., 2003). In both the cases of catchment disturbance described earlier, effects on IRES may be amplified by the periodic lack of flow. Thus, the accumulation of pollutants during cease-to-flow periods may generate significantly higher pollutant concentrations in IRES than those in perennial streams, and the periodic lack of connectivity can mean that most aquatic fauna cannot escape to less impacted sites.

Disturbances acting at these catchment scales can be extremely difficult to reverse, in part because of the sheer scale of the impacts and in part because there may be strong positive feedbacks reinforcing the degraded state. For example, to address catchment salinization, it is estimated that 40%–50% of a catchment must be revegetated. In itself, this is an ambitious target for large catchments, but can be almost impossible to achieve in low rainfall areas where the soils are heavily degraded (Ghassemi et al., 1995). In other cases, channel incision can greatly increase the shear stress associated with even small floods, exacerbating the problem of trying to restore natural channel morphology. However, there may

be isolated sources of degradation such as gully erosion contributing sediment to streams that can be efficiently targeted by restoration activities to reduce a catchment-scale impact (Hermoso et al., 2015). A consistent theme for ameliorating catchment-scale disturbances (as for others discussed below) is thus the need to undertake careful planning to decide on the most effective and efficient means of achieving restoration goals (Beechie et al., 2008).

5.4.5 HYDROLOGICAL IMPACTS AND THEIR RESTORATION

ALTERED FLOODING REGIMES

As with perennial streams, floods are a powerful shaping force in IRES. Large floods can reshape the channel, destroying and building new stream habitat, rejuvenating riparian vegetation, replenishing groundwater levels, and distributing plant propagules (e.g., Stromberg, 2001; Stromberg et al., 2007; Levick et al., 2008; Kondolf et al., 2013). Infrequent floods trigger riparian tree (e.g., cottonwood *Populus* spp.) recruitment in ephemeral sand-bed streams (Friedman and Lee, 2002), often with decadal intervals between floods and the accompanying recruitment. Similarly, large floods down IRES in the arid and semiarid lands of Australia promote recruitment of river red gum (*Eucalyptus camaldulensis*). For river red gums to remain healthy, flooding is needed every 3–4 years (Roberts and Marston, 2011; Catelotti et al., 2015). In both cases, recruitment can be greatly reduced by insufficient soil wetting, low groundwater levels, and grazing livestock. In IRES, the use of environmental flows released from dams to mimic natural floods could be a powerful restoration measure to restore riparian vegetation, but well-documented cases remain to be found.

The impacts of such extreme events, particularly floods, droughts, and wildfires, may be greatly magnified by human degradation of catchment condition. Extreme floods aided the creation of the arroyos in southwestern United States, cutting deep channels and removing riparian vegetation (Webb and Leake, 2006). Land uses such as intensive agriculture can exacerbate the impacts of floods. For example, a flash flood down a catchment which had been cleared and the channel straightened produced very heavy sediment loads and damaging erosive changes in channel configuration whereas in another catchment where intensive agriculture was declining and reforestation was extensive, the same flood event produced much lower sediment loads (Ortega et al., 2014). However, there was still severe erosion at sites with active channelization works that produced increased velocities and erosive energy.

LOSS OF FLOWS

Urbanization can threaten riparian zones by channelization (divorcing the stream from its riparian zone) and by clearing or fragmenting the riparian zone for the construction of buildings and infrastructure (Goodwin et al., 1997; Groffman et al., 2003). A further threat from urbanization and agricultural activities is the diversion of surface flows, reduced infiltration from impervious surfaces, and the extraction of groundwater, all of which may lower groundwater tables below the root zones of riparian plants (Stromberg et al., 1996, 2005; Webb and Leake, 2006) and contribute to increased flash flooding. The latter threat may favor the invasion of exotic plants (from ground cover to trees) which may supplant the native flora through competition and/or by taking advantage of the lowered groundwater tables (Lite and Stromberg, 2005; Richardson et al., 2007; Stromberg et al., 2007).

INCREASED PERIODS OF ZERO FLOWS

As dry periods are intrinsic to IRES, much of the biota has strategies to survive such events (Chapter 4.8); these strategies may even allow survival through severe suprasedseasonal droughts (Lake, 2011). From a restoration management point of view, both the timing of flows and spatiotemporal pattern of channel drying are critical. There are at least three distinctive spatial patterns of stream drying: the upstream channel is perennial whereas the downstream channel is intermittent and dries out either completely or to pools, the upstream channels are intermittent while downstream channels remain perennial, or the stream from its source to downstream reaches goes from being perennial to intermittent along its length (Lake, 2003; Chapter 2.3). Each of these patterns has different implications for the ability of individual taxa to access suitable refuge habitats. During nonflowing periods, headwater refuges may benefit very different taxa to those occurring downstream in refuge pools of lowland rivers (Chapter 4.8).

Frequent or prolonged drying of the channel may allow dry streambeds to be used by terrestrial animals to move, forage, or shelter, and colonized by terrestrial plants (Steward et al., 2012). Dry streambeds are also used by humans as a transport corridor and as venues for cultural activities. Changes in patterns of flow intermittence can arise from a number of causes. Farm dams or ponds, mainly for supplying water for livestock, are common on many dryland catchments. However, they can reduce stream flow and greatly extend the natural cease-to-flow period by acting as additional storage and evaporation basins (Callow and Smettem, 2009; Nathan and Lowe, 2012), with particularly damaging effects during and after droughts (e.g., O'Connor, 2001). The impact of farm dams on streamflow is especially critical during low rainfall seasons, capturing up to 30% of net runoff and greatly prolonging nonflowing periods, an effect that will be exacerbated by climate change in many areas (Fig. 5.4.3).

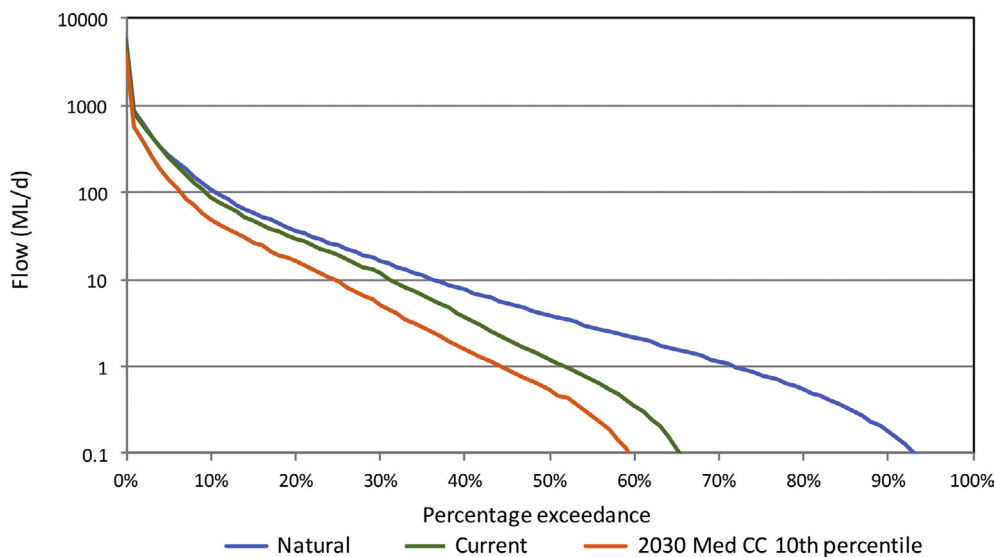


FIG. 5.4.3

Flow duration curves for Castle Ck, a second-order IRES in southeastern Australia. Individual curves represent the natural (blue line), current (green line), and projected future flows (orange line), highlighting the relative influences associated with farm dams (current) and climate change impacts (2030 med) on flow permanence.

BOX 5.4.1 ARTIFICIALLY INCREASED FLOW PERMANENCE IN IRES: EFFECTS AND RESTORATION CONSTRAINTS

The effects of artificially increased flow permanence, particularly when perennial flows replace previously intermittent ones, largely fall into three categories: (1) changes in the composition and abundance of flow-dependent biota; (2) associated changes in susceptibility to invasion by nonnative species; and (3) changes in water chemistry and biogeochemical cycling, especially at the soil-water interface. The lack of direct case studies means that these effects are simply inferred from comparisons between naturally perennial and naturally or artificially intermittent reaches. For example, most studies (Fritz and Dodds, 2005; Wood et al., 2005) report higher diversity and abundance of aquatic invertebrates in perennial streams. Similar trends have also been reported for fish (Rolls et al., 2012), and in most cases, there is some degree of nestedness (i.e., the fauna of intermittent sites is largely a subset of that from perennial sites (Dodds et al., 2004)).

It would be wrong, however, to conclude that perennialization leads to *positive* outcomes for the biota. For example, many plants and invertebrates have life history strategies that involve dormant seeds or eggs which hatch only after desiccation and rewetting (Boulton, 1989; Brock et al., 2003; Jenkins and Boulton, 2003; Chapter 4.8). Studies of wetlands subjected to increased hydroperturbations suggest that increasing flow permanence can act as a major disturbance in these systems (Golladay et al., 1997; Brock et al., 2003) by disrupting life cycles and modifying refuge habitats. In a restoration context, Bond et al. (2010) concluded that restoration of an intermittent flow regime would preferentially suit native fishes over introduced species that proliferated under artificially perennial flows. Conversely, naturally intermittent flows have been shown to facilitate persistence of native fishes in main stem and floodplain wetland habitats, respectively, due to the physiological constraints (e.g., temperature and DO tolerance) of some invasive species (McNeil and Closs, 2007). There are very few examples where naturally intermittent flows have been actively restored. Instead, there may be strong social pressures and, in some cases, conservation targets that promote establishing artificially perennial flows even where these preclude the reinstatement of the locally native ecosystems adapted to intermittent flows (Bateman et al., 2015).

ANTIDROUGHT

While increased intermittence from water extraction is perhaps the most widely considered form of hydrological disturbance affecting IRES (Chapter 5.1), the inverse situation may also arise when streams have higher levels of flows and flow permanence than would occur naturally (so-called antidrought; McMahon and Finlayson, 2003). Such inverse flow patterns are most likely to result from river regulation and the delivery of high flows for human use during the natural low-flow season, a common pattern in irrigation regions around the world (McMahon and Finlayson, 2003; Magilligan and Nislow, 2005; Magilligan et al., 2013). While some hydrological analyses demonstrate that irrigation releases can lead to increases in flow permanence and decreases in flow variability (Fernández et al., 2012; Alexandre et al., 2013), there are relatively few studies that specifically document the effects of increased flow permanence, particularly artificially induced perennial flows (perennialization, Chapter 5.1) on the structure and function of IRES. In part, this may reflect the underreporting of the effects of flow alteration and stream restoration on IRES; seldom have naturally intermittent flows been intentionally restored (Box 5.4.1).

5.4.6 RIPARIAN ZONE DEGRADATION AND RESTORATION

Riparian zones and their vegetation support a wide range of important ecological functions (Ewel et al., 2001). These include bank stabilization, provision of living and dead organic matter (both dissolved and particulate), and of habitat (including wood) for both aquatic and terrestrial biota, capture of sediment, retention and processing of nutrients, and moderation of extreme temperatures by shading (Ewel et al., 2001; Davies, 2010; Burt et al., 2013; Reich et al., 2016). As IRES mainly occur in arid and semiarid regions, the presence of surface water and groundwater fosters the development of distinct riparian zones replete with herbs, shrubs, and trees (Chapter 4.2), which provide key habitat and resources to a

rich fauna, from insects to mammals. This fauna can be quite different from and richer than the fauna in the surrounding resource-poor hinterland (Sabo et al., 2005; Johnson et al., 2007).

Given this importance, it is not surprising that restoration of IRES has concentrated on riparian zones rather than the much more difficult task of restoring catchments. Indeed, restoring riparian zones may be a way to quarantine streams from the impacts of disturbances in their catchments. Many disturbances such as the clearance of floodplains, livestock grazing, and the spread of exotic plants impact on riparian zones of IRES and may operate synergistically. The nature and variety of disturbances largely sets the agenda for restoration projects. Often, catchment clearing will decrease the width and tree density of riparian zones as well as fragmenting their continuity (Goodwin et al., 1997; Stella et al., 2013; Beller et al., 2016). If livestock are allowed access to the riparian zones and streambeds, vegetation can be lost through grazing, further compromising bank stability, and aquatic habitats may be destroyed by trampling and addition of excess nutrients by defecation (Fig. 5.4.4; Line et al., 2000; Steward et al., 2012; Reich et al., 2016). Furthermore, maintaining access for stock to drink from remnant pools is often seen as a factor motivating farmers to keep riparian zones along IRES unfenced.

Invasion of riparian zones of IRES by highly competitive plants, including trees and grasses, can be a major problem (Richardson et al., 2007). In southwestern arid United States, *Tamarix* (tamarisk, salt-cedar) and *Elaeagnus angustifolia* (Russian olive) are widespread and highly successful invaders of riparian zones along both intermittent and perennial streams (Friedman et al., 2005; Shafroth et al., 2005; Stromberg et al., 2009). Tamarisks readily invade riparian zones of IRES and there have been many projects involving millions of dollars to eradicate this plant (Shafroth et al., 2005; Shafroth and Briggs, 2008; Sher and Quigley, 2013). Tamarisk invasion and spread are substantially aided by a variety of anthropogenic disturbances, ranging from riparian vegetation clearing to regulated river flows. With its capacity to survive dry periods (including droughts), its high tolerance of soil salinity and its capacity to reach deep sources of groundwater, tamarisk may be a symptom of river degradation. Nevertheless, it also offers a means to prevent riparian zone erosion and provide shelter for terrestrial wildlife (Shafroth et al., 2005; Stromberg et al., 2007, 2009; Shafroth and Briggs, 2008; Sher and Quigley, 2013).



FIG. 5.4.4

A dry IRES in southeastern Australia showing the effects of stock access and overgrazing on bankside vegetation and morphology. Note the pugging of sediments in the channel from stock access during a moist phase.

Photo: Courtesy P. Reich.

Many IRES, some of considerable size, occur in arid to semiarid rangelands where production of livestock (especially sheep and cattle) is the major economic activity. In such areas, riparian zones offer shade, shelter from the heat, and access to water and forage (Kauffman and Krueger, 1984). Livestock having access to riparian zones can degrade water quality; alter stream channel morphology and riparian soils; and reduce instream, streambank, and riparian vegetation (Kauffman and Krueger, 1984; Goodwin et al., 1997; Belsky et al., 1999; Reich et al., 2016). Fencing off the riparian zone of streams to prevent livestock access can help restore both the stream and the riparian zone (Fig. 5.4.5; Kauffman and Krueger, 1984; Line et al., 2000; Kueper et al., 2003; Miller et al., 2010). In implementing fencing, it is important for the riparian zone to approach its pregrazing width (Hansen et al., 2015).

While there have been many livestock exclusion projects along IRES, published studies of riparian restoration by livestock exclusion are few (Reich et al., 2016). In a survey of fenced-off riparian zones along intermittent and perennial creeks, Burger et al. (2010) found that soil carbon concentrations increased with restoration and that soil nitrate and plant-available phosphorus were influenced by the concentrations in adjacent farmland. Sunohara et al. (2012) in a 5-year study on an IRES compared a 356-m length of riparian zone (3–5 m wide) with restricted cattle access with a 348-m riparian section downstream with cattle access. Excluding cattle from the riparian zone and channel of the IRES reduced mean loads of nutrients, total coliform bacteria, *Escherichia coli*, and *Enterococcus* spp. at both low and high flow conditions. Despite the short lengths of stream involved, these results indicate that intact riparian zones and restricted livestock access can reduce stream loads of nutrients and pathogenic bacteria even under the fluctuating flow conditions of IRES.

While fencing to exclude livestock and allowing natural regeneration of riparian vegetation is seen as an extremely cost-effective restoration strategy, several studies have demonstrated that the depletion of the original seed-bank coupled with the predominance of propagules of invasive plants, especially introduced pasture and weed species, may render this approach largely ineffective (Williams et al., 2008; O'Donnell et al., 2016). For example, in southeastern Australia, fencing off riparian zones and excluding livestock grazing along IRES resulted in an introduced pasture grass (*Phalaris aquatica*) rapidly coming to dominate the ground cover (Williams et al., 2008; Reich et al., 2016). Active restoration



FIG. 5.4.5

A drying IRES in southeastern Australia that has been fenced from stock and replanted. Note replantings in the background and significant growth of fringing vegetation.

Photo: Courtesy P. Reich.

involving physical and/or chemical removal of invaders and boosting native plant recruitment by adding seeds and/or seedlings may be necessary for effective recovery (Richardson et al., 2007). Many of the invading plants are nitrophilous and thus lowering the soil nitrogen levels may favor native plants. One approach that has been trialed effectively is the addition of freely available carbon (e.g., sugar) to riparian soils which increases nitrogen uptake by bacteria, thereby impeding the growth of invasive plants (e.g., Prober et al., 2005; Cole et al., 2016).

Riparian restoration of IRES may also restore terrestrial faunal assemblages. The restoration of bird assemblages appears to rely on successful restoration of native vegetation, together with the size of the restored area. Four years after the removal of cattle along perennial and intermittent sections of the San Pedro River in the San Pedro Riparian National Conservation Area, there was a substantial increase in riparian vegetation and abundance and species diversity of birds (Kueper et al., 2003). In contrast, after 8 years of riparian vegetation restoration, involving the removal of livestock and replanting along IRES in southern Australia (Fig. 5.4.6), recovery of the bird fauna was difficult to detect (Hale et al., 2015). These studies illustrate that the time for successful riparian restoration varies considerably, depending on the initial conditions, target indicators, and restoration goals.



FIG. 5.4.6

Aerial view of a site with riparian fencing and replanting along the banks of an IRES in southeastern Australia (see Reich et al. (2016) for details).

Photo: Courtesy P. Reich.

5.4.7 RESTORATION OF INSTREAM HABITAT AND REFUGES

For many degraded streams, restoration of instream habitat is one of the most commonly employed restoration strategies (Bernhardt et al., 2005, 2007). In a recent survey of restoration methods employed globally, “channel hydromorphic” and “instream hydromorphic” approaches accounted for 70% of restoration projects (Palmer et al., 2014). Such habitat restoration activities can occur at a number of scales.

At a local scale, habitat restoration most commonly focuses on the addition of habitat structure (e.g., rock, large woody debris) to provide cover and help restore local geomorphic diversity. For example in streams degraded by sedimentation, channelization, and removal of fallen timber, the addition of instream structures can increase hydraulic diversity and begin the process of restoring channel morphology over relatively short timescales. This action can thus complement the processes expected to unfold over decades following riparian restoration. Such methods have been shown to be effective in IRES but their effectiveness may wane during severe dry periods. In a survey of two sand-slugged IRES, Bond and Lake (2003b) reported that fish were mostly found around structurally complex habitat, notably that created by large wood. In an attempt to restore habitat and both fish and invertebrate assemblages, wood structures (railway sleepers) were positioned at nine sites in each creek devoid of existing habitat structure. Fish surveys occurred before and after wood installation. In the short term, there were increases in abundance, notably of the galaxiid fish *Galaxias olidus* at the wood structures (Bond and Lake, 2005). However, during a period of extreme drought, fish disappeared from these restored reaches, but persisted within remnant refugial pools (Bond, 2012). As sand sedimentation has greatly reduced the availability of these pools, and as the drought was particularly severe (10 years), subsequent recolonization was predicted to be slow (Bond, 2012).

At larger spatial scales, there have been efforts to restore the reach-scale channel form by constructing stable, possibly meandering, channels (Kondolf, 2012; Kondolf et al., 2013). However, given the tendency for periodic large floods in many IRES, such “stable” channels are prone to destruction. Instead, restoration should promote channels capable of evolving in response to these episodic floods (Kondolf et al., 2013; Ortega et al., 2014; Beagle et al., 2016). Restoration of this flood-dependent nature involves restoring zones both longitudinally and laterally where the river in flood “can flood, erode, deposit, and shift course without conflicting with human uses” (Kondolf et al., 2013). Such a zone has been called the channel migration zone (Beagle et al., 2016), “territorio fluvial” (Ollero and Ibasate, 2012) or “espace de liberté” (Kondolf, 2012). As Kondolf (2012, p. 239) says, “The most effective approach to restoring rivers will often be for us to stand aside, and give the river its space.” Large episodic floods in IRES are a vivid example of the power of extreme events (Reich and Lake, 2015), although their restorative powers may be limited in cleared catchments.

5.4.8 RESTORING REFUGES AND COLONIZATION PATHWAYS

IRES are, by their very nature, highly variable systems with disturbances at both ends of the flow regime ranging from seasonal and suprasedasonal droughts to floods (Chapters 2.2 and 2.3). Although native biota are typically adapted to persist over multiple generations in these environments, the ability to move between breeding, residential, and refuge habitat areas can be critical (Bond and Lake, 2003a). Thus, as well as restoring permanent refuges, it is necessary to restore subsequent postdisturbance recolonization pathways (connectivity). If not, then restoration is unlikely to succeed (Bond et al., 2008; Robson et al., 2013).

Refuge habitats are used by a wide spectrum of organisms from attached algae to micro- and macroinvertebrates and fish (Boulton and Lake, 2008; Robson et al., 2008, 2013). In IRES restoration, provision of persistent and drought-resistant refuge habitats is a critical challenge (Bond et al., 2008). Restoration of refuge habitats can be compromised in sand-bed IRES by the mobility of the streambed on the falling limb of the hydrograph, which infills pools prior to the loss of flow.

Restoring connectivity as a means of facilitating recovery following drying events can also be critical. For example, Dexter et al. (2014) demonstrated the need for hydrological connectivity to allow fish (*Galaxias olidus*) to access breeding sites within headwater reaches of a lowland stream network. This only occurs during nondrought years, when flows persist throughout the breeding season. In very dry years, access to breeding habitats is likely blocked (Dexter et al., 2014). Given that *G. olidus* live for only 3–4 years, relatively frequent connections are critical to their population persistence. Furthermore, to achieve hydrological connectivity, removal of artificial barriers may be an important restoration activity. In IRES, the predominance of low flows can mean that even small barriers, such as road pipes and culverts, can prevent recolonization following the recurrence of stream flows.

One aspect of connectivity that warrants consideration is the connection between multiple small-scale restoration projects distributed throughout the catchment (Kondolf et al., 2008). If restoration can only proceed at the local scale and projects are incrementally begun, then at the catchment level, a program of restoration should be devised that maximizes ecological benefits as sites are sequentially added. This raises the question: if restoration projects are at small spatial extents, are there ways by which their spatial distribution can be implemented to provide progressively greater restoration benefits at larger spatial extents? This question appears to be unresolved, and yet it remains pertinent to IRES restoration (Table 5.4.1).

5.4.9 MEASURING RESTORATION TRAJECTORY

The selection of indicators to assess the success of restoration projects can be a challenge, especially in IRES. To identify salient indicators for loss of connectivity and habitat fragmentation in an IRES, the Odeouca in Portugal, prior to the completion of a large dam, Hughes et al. (2010) assessed a range of potential indicators from macrophytes to birds. The most appropriate indicators of future flow changes were benthic macroinvertebrates and endemic fish. In many streams, changes in flow from perennial to intermittent and vice versa rely on the groundwater levels. One example is the San Pedro River in Arizona. Changes in groundwater level in this IRES are reflected in riparian vegetation composition with herbs reacting more rapidly than trees to groundwater changes (Stromberg et al., 2006, 2013). In perennial sections, native trees are dominant but with declining groundwater levels, *Tamarix*, tolerant of the low groundwater levels, may invade.

The importance of monitoring ecosystem processes in restoration has become increasingly recognized (Dolédec and Bonada, 2013; Boudell et al., 2015) along with the monitoring of ecosystem services (Palmer et al., 2014). The selection of indicators should consider the prevailing hydrological context (flowing, nonflowing and dry) as well as the timing (i.e., season). Understanding how each indicator responds to antecedent or prevailing hydrological states (e.g., time since last state, duration of current state, Table 5.4.1) is also likely to be important (Chapter 5.1).

5.4.10 THINGS TO CONSIDER FOR NOW AND FOR THE FUTURE

IRES and their catchments experience a wide variety of human disturbances varying in spatiotemporal extent and severity. As many IRES occur in dryland regions and their catchments are often of comparatively low economic value with low human population density, concern for their protection and conservation has lagged behind that directed toward more productive and populated areas. Consequently, human disturbances through land clearing, livestock grazing, low-yield grain production, and mining may be extensive and, in turn, impact severely on the morphology and ecology of dryland IRES.

Compared with restoration of perennial streams, concern for restoration of IRES appears limited, possibly due to their low perceived economic and social values (Chapter 5.2). A further difficulty in attempting to restore IRES lies in their very nature with variable wet and dry periods. This makes setting instream restoration targets an awkward exercise due to the very dynamic ecological changes which occur both when water in the channel is flowing, standing, or absent. Perhaps this explains the finding that, judging from the literature, the most common form of restoration in IRES is riparian zone revegetation.

Successful restoration of IRES is most likely to be achieved by adopting a catchment-wide approach. This involves identifying and assessing the spatial extent of legacy and current disturbances, and implementing interventions to curb or eliminate the disturbances. With lower human populations and less intensive forms of exploitation, it may be less challenging to execute restoration measures at the catchment scale in IRES than in perennial streams.

Successful restoration over a range of spatial extents is tightly linked to the length of time required to reach the set targets. Some targets, such as ground cover in riparian restoration, may be reached in a relatively short period while others such as tree maturity and channel reshaping may take many years, if not decades. Most stream restoration projects take substantial time yet funding and monitoring are rarely supported up to completion. This problem of limited funding is common and may be more severe for IRES restoration as many of these streams are not highly valued (Chapter 5.2) and restoration is slow, being governed by the periodicity of flow.

Most restoration measures can be described as “reactive”—reacting to legacy and current disturbances (Palmer et al., 2008; Bain et al., 2012). However, restoration can be planned and implemented in a proactive way (Palmer et al., 2008), which involves implementing measures now which may curb the forces of future disturbances. Given current and projected declines in rainfall and runoff due to climate change in dryland and Mediterranean regions (Jaeger et al., 2014; Pumo et al., 2016), IRES in these regions can also be expected to flow less often, with reduced flows and less groundwater recharge (Meixner et al., 2016). Thus, restoration could involve proactive measures such as the implementation of environmental flows (Palmer et al., 2009), increasing the effectiveness of refuges (e.g., pools) in streams, reducing water capture and extraction in catchments (e.g., decommissioning farm dams), increasing connectivity (e.g., dam removal), and restoring riparian vegetation (Davies, 2010; Stromberg et al., 2013).

Restoration goals in IRES usually focus on the flowing phase. However, nonflowing or dry streambeds are an essential component in the ecological dynamics of IRES (Steward et al., 2012). They can be degraded by human activities such as rubbish dumping, livestock access and grazing, gravel/sand extraction, and input of wastewater and sewage. Thus, in many cases it is essential that restoration actions in IRES consider both the wet (flowing and nonflowing) and dry streambed conditions.

5.4.11 CONCLUSIONS

A great number and variety of IRES with a wide range of flow patterns occur globally. Judging from the literature, restoration activities appear much rarer in IRES than in perennial streams and their catchments. From local sites to catchments, IRES host a wide range of human disturbances operating over a broad range of spatiotemporal scales and which appear to be less well understood than those operating in perennial systems. Ecological restoration of IRES, as for all restoration projects, needs to be carefully planned with a schedule of successive steps essential to produce ecologically beneficial outcomes.

Ideally, restoration of IRES should be planned and implemented at the catchment scale. However, this rarely occurs. There have been two forms of restoration at this scale: altering catchment land management and large-scale channel restoration. The major form of IRES restoration reported in the literature focuses on riparian zone restoration, with the major activity being revegetation. To be successful, revegetation takes time (years-decades), a requirement which is seldom funded and achieved. At local scales such as sites and reaches, adding habitat structure and bolstering dry-period refuges have been the major restoration activities. Achieving the latter requires careful attention to stream hydrology, including groundwater interactions as well as geomorphic attributes. Where restoration occurs via multiple small-scale restoration projects, such projects must be planned in a way that maximizes the benefits at the catchment scale. This requires prioritizing the types and location of various restoration activities (Beechie et al., 2008).

Finally, there remains a strong need for further monitoring of well-designed restoration projects focused on IRES. Not only must monitoring treat these as long-term projects (rather than as the short-term inadequate and incomplete projects that now prevail), monitoring should occur only where restoration activities have been well designed and planned with specific hypotheses and timelines regarding the expected biophysical response trajectories.

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